

Suvrajit Bhattacharjee

Postdoctoral fellow

Contact information

- Wydział Matematyki (Sekretariat 2009),
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Poland.
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Research interests

- Operator algebras, quantum groups and noncommutative geometry

Employment

- Faculty of Mathematics, University of Białystok, Białystok, Poland,
Postdoc,
July 2025 – present.
- Indian Statistical Institute, Kolkata, India,
Visiting Scientist,
April 2025 – June 2025.
- Mathematics Institute, University of Oslo, Oslo, Norway,
Postdoc,
April 2023 – March 2025.
- Mathematical Institute of Charles University, Prague, Czech Republic,
Postdoc,
April 2022 – March 2023.
- Indian Institute of Science Education and Research, Kolkata, India,
National Postdoctoral Fellowship, Science and Engineering Research Board,
India,
December 2021 – February 2022.
- Indian Statistical Institute, Kolkata, India,
Visiting Scientist,
January 2021 – November 2021.
- Institute of Mathematics, Polish Academy of Sciences, Warsaw, Poland,
Adjunct Professor, (Postdoc),
October 2020 – September 2022 (couldn't join due to travel restrictions
caused by the pandemic).

Education

- Indian Statistical Institute, Kolkata
Ph.D. in Mathematics, December 2020
Dissertation topic: Quantum symmetries in noncommutative geometry
Advisor: Prof. Debashish Goswami
- Ramakrishna Mission Vivekananda University
M.Sc. in Mathematics, June 2015
CGPA: 9.06
- University of Calcutta
B.Sc. in Mathematics, June 2013
First Class Honors

Publications and preprints

- with Forough, M., “Quasi-invariant lifts of completely positive maps for groupoid actions”, arXiv preprint math.OA/2405.07859, 2024.
- with Joardar, S., “Equivariant C^* -correspondences and compact quantum group actions on Pimsner algebras”, Canad. J. Math., Vol. 77, 2025.
- with Goswami, D., “Complex structures on Three-point space”, Proceedings of the International Conference on Infinite Dimensional Analysis, Quantum Probability and Related Topics, QP38, 2024.
- with Joardar, S., Roy, S., “Braided quantum symmetries of graph C^* -algebras”, Internat. J. Math., Vol. 35, 2024.
- with Anshu, Rahaman, A., Roy, S., “Anyonic quantum symmetries of finite spaces”, Lett. Math. Phys., Vol. 113, 2023.
- with Joardar, S., Mukhopadhyay, S., “Levi-Civita connections from toral actions”, J. Geom. Phys., Vol. 181, 2022.
- with Chirvasitu, A., Goswami, D., “Quantum Galois groups of subfactors”, Internat. J. Math., Vol. 33, 2022.
- with Biswas, I., Goswami, D., “Generalized symmetry in noncommutative (complex) geometry”, J. Geom. Phys., Vol. 166, 2021.
- with Goswami, D., “Hopf coactions on odd spheres”, J. Algebra, Vol. 539, 2019.

Honors and awards

- Fall Semester 2023, Fields Institute Thematic Program in Operator Algebras, Postdoctoral Fellowship.
- 2021–2023, Science and Engineering Research Board, India, National Postdoctoral Fellowship.

- 2021–2023, National Board for Higher Mathematics, Postdoctoral Fellowship.
- 2021–2023, Indian Institute of Science Education and Research, Postdoctoral Fellowship.
- 2015–2020, Indian Statistical Institute, Ph.D. Fellowship.
- 2015–2020, National Board for Higher Mathematics, Ph.D. Scholarship.
- 2015–2020, Council of Scientific and Industrial Research, Ph.D. Scholarship.
- 2013–2015, National Board for Higher Mathematics, M.Sc. Scholarship.

Talks

- Crystallization and K-theoretical aspects of the quantum twistor bundle, Noncommutativity behind the dunes, Delft University of Technology, Delft. (October 2024)
- Braided quantum groups and their actions, Noncommutativity along the North sea in Odense, University of Southern Denmark, Odense. (August 2024)
- Equivariant lifting problem for continuous fields of completely positive maps, Operator Algebra Seminar, University of Oslo, Oslo. (September 2023)
- Equivariant lifting problem for continuous fields of completely positive maps, Quantum Groups and Noncommutative Geometry in Prague, Charles University, Prague. (September 2023)
- (Equivariant) lifting problem for completely positive maps, Indian Statistical Institute, Kolkata. (July 2023)
- Actions of compact quantum groups 2, NCG&T Malý Seminar, Prague. (November 2022)
- Actions of compact quantum groups 1, NCG&T Malý Seminar, Prague. (November 2022)
- An introduction to (braided) quantum groups, Inter IISER-NISER Math meet, Indian Institute of Science Education and Research, Kolkata. (June 2022)
- Kasparov product and its properties, NCG&T Malý Seminar, Prague. (April 2022)
- Braided quantum symmetries of graph C^* -algebras, NCG&T Seminar, Prague. (April 2022)
- Kasparov modules and constructions, NCG&T Malý Seminar, Prague. (April 2022)
- Braided quantum symmetries of graph C^* -algebras, Quantum Groups Seminar, Copenhagen. (March 2022)

- Quantum Groups, Indian Institute of Science Education and Research, Kolkata. (February 2020)
- Hopf Algebroids in Noncommutative Geometry, Noncommutative Geometry and Its Applications @ NISER, National Institute of Science Education and Research, Bhubaneswar. (January 2020)
- Generalized Symmetry in Noncommutative Complex Geometry, Quantum flag manifolds, Charles University, Prague. (September 2019)
- Hopf Algebroids, Graduate seminar, Indian Statistical Institute, Kolkata. (April 2019)
- Quantum Symmetry, Graduate seminar, Indian Statistical Institute, Kolkata. (April 2018)
- Hopf Coactions on Commutative Algebras, Graduate seminar, Indian Statistical Institute, Kolkata. (March 2017)
- Algebraic Groups and Hopf Algebras, Graduate seminar, Indian Statistical Institute, Kolkata. (September 2016)
- Infinite Galois Theory, Graduate seminar, Indian Statistical Institute, Kolkata. (January 2016)

Events

- Co-organizer of the noncommutative geometry session of IWOTA 2025, University of Twente, Enschede. (July 2025)
- Co-organizer of the workshop Geometry and Analysis of Quantum Groups 2023, University of Oslo, Oslo. (November 2023)

Workshops and conferences attended

- Noncommutativity along the North sea in Odense, University of Southern Denmark, Odense. (August 2024)
- Noncommutativity in the North - MikaelFest, University of Gothenburg – Chalmers University of Technology, Gothenburg. (June 2024)
- Point sets: Dynamics, Sampling, and Operator Algebras, University of Oslo, Oslo. (May 2024)
- Quantum Groups and Noncommutative Geometry in Prague, Charles University, Prague. (September 2023)
- Operator Algebras and Mathematical Physics (Yasu Festa 60), University of Tokyo, Tokyo. (July 2023)
- Quantum Groups: Current Trends and New Perspectives, University of Oslo, Oslo. (December 2022)
- Noncommutative Harmonic Analysis and Quantum Groups, Institute of Mathematics, Polish Academy of Sciences, Będlewo. (September 2022)

- Noncommutative Geometry and Its Applications @ NISER, National Institute of Science Education and Research, Bhubaneswar. (January 2020)
- KBS Fest, Indian Statistical Institute, Bangalore. (December 2019)
- Quantum Flag Manifolds, Charles University, Prague. (September 2019)
- Recent Advances in Operator Theory and Operator Algebras, Indian Statistical Institute, Bangalore. (December 2018)
- International Conference on Non-commutative Geometry, S. N. Bose National Centre for Basic Sciences, Kolkata. (November 2018)
- Conference on Quantum Groups and Non-commutative Geometry, National Institute of Science Education and Research, Bhubaneswar. (January 2018)
- CIMPA Research School on Recent Trends in Non-commutative Algebras, Indian Institute of Science Education and Research, Pune. (June 2017)

Academic visits

- Prof. Michal Doucha, Institute of Mathematics of Czech Academy of Sciences, Prague. (February 2025)
- Prof. Debashish Goswami, Indian Statistical Institute, Kolkata. (July 2023)
- Prof. Adam Skalski, Institute of Mathematics, Polish Academy of Sciences, Warsaw. (August 2022)
- Dr. Réamonn Ó Buachalla, Mathematical Institute, Charles University, Prague. (March 2022)
- Dr. Soumalya Joardar, Indian Institute of Science Education and Research, Kolkata. (February 2020)
- Dr. Sutanu Roy, National Institute of Science Education and Research, Bhubaneswar. (April, 2018; November 2019)
- Prof. Arup Pal, Indian Statistical Institute, Delhi. (January 2018)
- Prof. Indranil Biswas, Tata Institute of Fundamental Research, Mumbai. (November 2017)

Teaching experience

- Fall 2019, Teaching Assistant, Riemannian Geometry.
- Fall 2018, Teaching Assistant, Linear Algebra.
- Spring 2018, Teaching Assistant, Lie Groups and Lie Algebras.

References

- Prof. Debashish Goswami,
Stat Math Unit, Indian Statistical Institute,
203 B.T. Road, Kolkata - 700108, India,
`goswamid@isical.ac.in`
- Prof. Sergey Neshveyev,
Matematisk institutt, Universitetet i Oslo,
P.O. Box 1053, Blindern, 0316 Oslo, Norway,
`sergeyn@math.uio.no`
- Dr. Réamonn Ó Buachalla,
Mathematical Institute, Charles University,
Sokolovská 49/83, 18 675 Prague 8, Czech Republic,
`obuachalla@karlin.mff.cuni.cz`
- Dr. Karen R. Strung,
Institute of Mathematics, Czech Academy of Sciences,
Žitná 25, 115 67 Prague 1, Czech Republic,
`strung@math.cas.cz`